Task 1 — Pre-Lab Worksheet

Given formula:

$$T_{\text{overflow}} = \frac{(\text{PSC}+1)\times(\text{ARR}+1)}{f_{\text{TIM\_CLK}}}$$

Q1. TIM6, $f = 90$ MHz

1 ms:

$$(\text{PSC}+1)\times(\text{ARR}+1) = 0.001\times 90\times 10^6 = 90000$$

PSC = 89, ARR = 999 $\quad\Rightarrow\quad$ $90\times 1000 = 90000$ $\quad$ Actual = 1 ms

Also valid: PSC = 899, ARR = 99 $\Rightarrow$ $900\times 100 = 90000$

100 μs:

$$(\text{PSC}+1)\times(\text{ARR}+1) = 100\times 10^{-6}\times 90\times 10^6 = 9000$$

PSC = 8, ARR = 999 $\quad\Rightarrow\quad$ $9\times 1000 = 9000$ $\quad$ Actual = 100 μs

Also valid: PSC = 89, ARR = 99 $\Rightarrow$ $90\times 100 = 9000$

500 ms:

$$(\text{PSC}+1)\times(\text{ARR}+1) = 0.5\times 90\times 10^6 = 45000000$$

PSC = 4499, ARR = 9999 $\quad\Rightarrow\quad$ $4500\times 10000 = 45000000$ $\quad$ Actual = 500 ms

Also valid: PSC = 8999, ARR = 4999 $\Rightarrow$ $9000\times 5000 = 45000000$

1 s:

$$(\text{PSC}+1)\times(\text{ARR}+1) = 1\times 90\times 10^6 = 90000000$$

PSC = 8999, ARR = 9999 $\quad\Rightarrow\quad$ $9000\times 10000 = 90000000$ $\quad$ Actual = 1 s

Also valid: PSC = 17999, ARR = 4999 $\Rightarrow$ $18000\times 5000 = 90000000$

Q2. TIM3 PWM at 1 kHz, $f = 90$ MHz

(a)

$$(\text{PSC}+1)\times(\text{ARR}+1) = \frac{90\times 10^6}{1000} = 90000$$

Choose PSC = 89, ARR = 999 $\quad\Rightarrow\quad$ $90\times 1000 = 90000$ $\quad$ $f_{\text{PWM}} = 1$ kHz

(b) $\quad$ $\text{CCR1} = D\times(\text{ARR}+1) = D\times 1000$

$$\begin{aligned}
25\% :&\quad \text{CCR1} = 0.25\times 1000 = 250\\
50\% :&\quad \text{CCR1} = 0.50\times 1000 = 500\\
75\% :&\quad \text{CCR1} = 0.75\times 1000 = 750\\
100\% :&\quad \text{CCR1} = 1.00\times 1000 = 1000
\end{aligned}$$

(c)

$$\text{CCR1} = D\times(\text{ARR}+1)$$

Q3. TIM1 for WS2812B, $f = 180$ MHz, PSC $= 0$

$T_{\text{clk}} = \dfrac{1}{180 \times 10^6} \approx 5.556$ ns

(a) ARR for $T_{\text{bit}} = 1250$ ns:

$$\text{ARR} + 1 = 1250 \times 10^{-9} \times 180 \times 10^6 = 225 \quad \Rightarrow \quad \text{ARR} = 224$$

(b) CCR1 for logic-1 ($T_{1H} = 800$ ns):

$$\text{CCR1} = 800 \times 10^{-9} \times 180 \times 10^6 = 144$$

(c) CCR1 for logic-0 ($T_{0H} = 400$ ns):

$$\text{CCR1} = 400 \times 10^{-9} \times 180 \times 10^6 = 72$$

(d) Tolerance check (tolerance $= \pm 150$ ns):

$T_{1H} = 144 \times 5.556 = 800$ ns (nominal 800 ns)

$T_{0H} = 72 \times 5.556 = 400$ ns (nominal 400 ns)

$T_{0L} = 1250 - 400 = 850$ ns (nominal 850 ns)

$T_{1L} = 1250 - 800 = 450$ ns (nominal 450 ns)

All values are within ± 150 ns tolerance.

Q4. DWT Cycle Counter, $f_{\text{CPU}} = 180$ MHz

$$t = \frac{\text{cycles}}{f_{\text{CPU}}} = \frac{\text{cycles}}{180 \times 10^6}$$

$$3600: \quad \frac{3600}{180 \times 10^6} = 20000 \text{ ns} = 20\ \mu\text{s} = 0.02 \text{ ms}$$

$$90000: \quad \frac{90000}{180 \times 10^6} = 500000 \text{ ns} = 500\ \mu\text{s} = 0.5 \text{ ms}$$

$$9000000: \quad \frac{9000000}{180 \times 10^6} = 50000000 \text{ ns} = 50000\ \mu\text{s} = 50 \text{ ms}$$